

4 Sequences and series	4.1	Understand and use the binomial expansion of $(a + bx)^n$ for positive integer n; the notations $n!$ and nC_r link to binomial probabilities. Extend to any rational n, including its use for approximation; be aware that the expansion is valid for $\left \frac{bx}{a} \right < 1 \text{ (proof not required)}$	Use of Pascal's triangle. Relation between binomial coefficients. Also be aware of alternative notations such as $\binom{n}{r}$ and nC_r Considered further in Paper 3 Section 4.1. May be used with the expansion of rational functions by decomposition into partial fractions May be asked to comment on the range of validity.
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Topics	What students need to learn:		
	Content	Guidance	
4 Sequences and series <i>continued</i>	4.2	Work with sequences including those given by a formula for the nth term and those generated by a simple relation of the form $x_{n+1} = f(x_n)$; increasing sequences; decreasing sequences; periodic sequences.	For example $u_n = \frac{1}{3n+1}$ describes a decreasing sequence as $u_{n+1} < u_n$ for all integer n $u_n = 2^n$ is an increasing sequence as $u_{n+1} > u_n$ for all integer n $u_{n+1} = \frac{1}{u_n}$ for $n > 1$ and $u_1 = 3$ describes a periodic sequence of order 2
	4.3	Understand and use sigma notation for sums of series.	Knowledge that $\sum_1^n 1 = n$ is expected
	4.4	Understand and work with arithmetic sequences and series, including the formulae for n th term and the sum to n terms	The proof of the sum formula for an arithmetic sequence should be known including the formula for the sum of the first n natural numbers.
	4.5	Understand and work with geometric sequences and series, including the formulae for the n th term and the sum of a finite geometric series; the sum to infinity of a convergent geometric series, including the use of $ r < 1$; modulus notation	The proof of the sum formula should be known. Given the sum of a series students should be able to use logs to find the value of n. The sum to infinity may be expressed as S_∞
	4.6	Use sequences and series in modelling.	Examples could include amounts paid into saving schemes, increasing by the same amount (arithmetic) or by the same percentage (geometric) or could include other series defined by a formula or a relation.